

DURHAM UNIVERSITY

MCMC Convergence and Tessellation Budget Sensitivity in the AddiVortes Model

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September 8, 2026

1 Introduction

The Additive Voronoi Tessellations (AddiVortes) model is a multivariate regression model that uses Voronoi tessellations to partition the covariate space in an additive ensemble model. Unlike other partition methods, such as decision trees, this has the benefit of allowing the boundaries of the partitions to be non-orthogonal and nonparallel to the covariate axes. The AddiVortes model uses a similar sum-of-tessellations approach and a Bayesian backfitting MCMC algorithm to the BART model. We use regularization priors to limit the strength of individual tessellations and accepts new models based on a likelihood. [1]

Two lesser understood aspects of the AddiVortes model are MCMC convergence and budget sensitivity. We want to be able to better understand MCMC convergence behaviour as the dimensionality of the problem (the number of input covariates), and the sample size grow. We also want to understand the sensitivity of the predictive accuracy of the model to changes in budget parameters, which can dramatically affect the compute time it takes to fit the model.

In this 8-week project, funded by N8 CIR through their undergraduate internship programme, I investigated these properties of the model, making use of the Hamilton HPC Service of Durham University. I worked directly with the existing codebases for the Python and R implementations of the algorithm, collaborating with my project supervisor, Prof. John Paul Gosling, to make optimisations which have had a direct impact on the speed of model fitting, as well as introducing features to aide the exploration of the key project aims, specifically convergence metrics.

2 Model parameters

In order to explore the trade off between model fitting time and model accuracy in the model, we first need to understand the choices we make when fitting the model. These are the numerical parameters such as the number of tessellations to use. A summary of the parameters we will investigate is below. Further details of the role these parameters play in model fitting can be found in [1]. The search ranges given here are based on the current work of Adam Stone.

Choice/parameter	Symb	Default value	Search range
Number of tessellations	m	200	[20, 500]
Noise degrees of freedom hyperparam	ν	6	[1, 10]
Noise alignment hyperparam	q	0.85	[0.6, 0.999]
Number of covariates hyperparam	ω	$\min(3, p)$	[1, 7] (depends on p)
Number of centers hyperparam	λ	25	[1, 50]
Number of MCMC iterations	—	1200	[500, 10000]
Number of MCMC burn-ins	—	200	[0.01, 0.5] (as a proportion)

Table 1: Model parameters

3 Benchmark analysis and subsequent code changes

Before starting work on the core aims of exploring MCMC convergence and budget sensitivity, we first needed to compare the 2 current implementations of the AddiVortes model. We compared fit times, prediction times, and in-sample and out-of-sample RMSEs between the Python and R implementations. The purpose of this was twofold. Firstly, we were able to compare specific results for large discrepancies which could indicate bugs in one of the two implementations. Secondly, we carried out an analysis of the results to determine which of the implementations would be ideal for further use, that is in exploring convergence and budget sensitivity.

The code used to generate the benchmarking data, as well as the generated data itself and all referenced files, are available to view here [2]. This data was generated using the ADDIVORTES R package [3], and the ADDIVORTES Python package. [4].

3.1 Benchmarking methodology

To ensure fairness, consistency, and accuracy with these tests, the following controls were implemented on these benchmarks:

1. The benchmarks were run on a new GitHub Codespace, a clean installation of Python and R were used, with only the necessary packages downloaded to minimise background resource usage. The Codespace used is a 2-core 2.8GHz system with 8GB of memory. Exact details of hardware used can be found at `benchmarks\mySpecs.html` [2].
2. The code was set up such that each implementation used only a single thread, to counteract the potentially different methods of multi threading between the two languages. This choice was made as we are working under the assumption that when we are working with HPC resources, the code will be optimised to allow multiple threads to be used at once. This choice allows us to compare algorithmic efficiency in each implementation more directly.
3. The same models and predictions were fitted repeatedly in order to capture the small variations in fit and prediction times between runs. This will also produce a set of slightly varying prediction results due to the random nature of the MCMC fitting algorithm, and allow us to better analyse how consistent our results are.
4. The benchmarks were run sequentially rather than simultaneously to minimise the risk of shared resources skewing the results, where one script could take computational priority over another, leading to a larger discrepancy in results.
5. The timings were taken only between the computation required for fitting and generating predictions, with the pre-generated datasets already loaded in prior to this.

3.2 Datasets used in benchmarking

To allow for a wide scope of comparison between the two implementations, the following 4 different datasets were used in benchmarking. These 4 datasets were all loaded/generated as described using R, then split into test and train sets for the input covariates and exported to the files `xtrain.csv`, `ytrain.csv`, `xtest.csv`, and `ytest.csv` for each dataset. The code used to do this can be found at `benchmarks\dataset-export.R` [2]. These fixed datasets were then used in all the following benchmarks.

3.2.1 Boston dataset

The Boston Housing Dataset, derived from information from the U.S. Census Service. This consists of 13 numerical covariates, and we model it using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`

3.2.2 Synthetic dataset

We have constructed a synthetic dataset using noisy functions of independently and identically distributed $U[0,1]$ observations. This consists of 5 numerical covariates, and we model it by taking all the parameters as their default values.

3.2.3 Spherical dataset

We have constructed random longitudes and latitudes to represent positions on a sphere, and a noisy response function. This consists of 2 numerical covariates (the coordinates), and we model it

using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`, `totalMCMCiter = 500`, `mcmcBurnIn = 100`, `metric = "S"`. Note that this uses great circle distance, the shortest distance between two points over the surface of the sphere.

3.2.4 Categorical dataset

We have constructed this dataset using random numerical and categorical covariates, as well as a noisy response function. This consists of 2 numerical and 2 categorical covariates, and we model it using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`, `totalMCMCiter = 500`, `mcmcBurnIn = 100`, `catScaling = 1`

3.3 Benchmark iterations

The benchmarks produced model fits on the training data and then used the model fits to produce predictions on the training and test data. The code can be found at `benchmarks\tests.R` and `benchmarks\tests.py` [2]. Full details of the benchmarking results can be found at `benchmarks\benchmark-report.pdf` [2]. Here I will summarise the results of each iteration of running these benchmarks, the findings of the results, and the changes that were made to the code at each iteration which led to the subsequent improvements in performance. For the sake of brevity, we compare the four models using the unweighted mean of their mean fit and prediction times, that is, we average the four per-model means directly, regardless of how many runs each model comprises. We will do the same for their in-sample and out-of-sample RMSEs. In order to not skew the summary results with data on different scales, the percentage differences reported here are computed as the average of the four individual test-level percentage differences, rather than recalculated from the aggregated mean values.

3.3.1 Iteration 1

The benchmarks were initially run for 106 iterations each on version 0.6.5 of the two implementations. The summary results are below:

Metric	Mean
Fit Mean (R)	5.161
Fit Mean (Python)	4.460
Fit % Diff	-16.842
Pred Mean (R)	3.870
Pred Mean (Python)	1.851
Pred % Diff	-50.844
iRMSE Mean (R)	1.315
iRMSE Mean (Python)	1.339
iRMSE % Diff	2.257
oRMSE Mean (R)	3.153
oRMSE Mean (Python)	2.590
oRMSE % Diff	-11.801

Table 2: Summary results for iteration 1 benchmarking

The cause of the 11.8% mean decrease in oRMSE in Python over R is an outlier in the benchmarking data, which is that for test 3 (the only test using spherical metric), there is a 51% decrease in oRMSE when using Python over R. More strangely, there is no comparable decrease in iRMSE in this test. This could suggest there is some error in the predict method of one of the implementations, specifically that the setting of using spherical metric is not being properly stored or applied. On further investigation, it was found that this error did occur in the R implementation of the algorithm, and a fix was implemented.

Additionally, at this point a change was made to the R implementation. The nearest neighbour algorithm used was slow and was calculating and storing many more distances than were required for the algorithm, where we only ever need the single nearest neighbour. The algorithm was adapted to instead just find the single nearest neighbour through a linear scan.

Both of these changes were made and the version numbers for both implementations were bumped to 0.6.6

3.3.2 Iteration 2

The benchmarks were now run for 500 iterations each on version 0.6.6 of the two implementations. The summary results are below:

Metric	Mean
Fit Mean (R)	4.124
Fit Mean (Python)	3.969
Fit % Diff	-8.832
Pred Mean (R)	2.699
Pred Mean (Python)	1.670
Pred % Diff	-37.703
iRMSE Mean (R)	1.314
iRMSE Mean (Python)	1.340
iRMSE % Diff	2.499
oRMSE Mean (R)	2.568
oRMSE Mean (Python)	2.588
oRMSE % Diff	1.286

Table 3: Summary results for iteration 2 benchmarking

The first thing to note is that the discrepancy in results for test 3 was fixed in this iteration, with results now being roughly in line. From the results, we can see that Python has faster mean fit times and much faster mean prediction times, and whilst the iRMSE and oRMSE means are slightly higher, they are quite comparable to the R implementation. Hence, we decided to select the Python implementation for further use.

4 Convergence of model fits

In order to understand how stable our predictions based on an AddiVortes fit will be, and the mathematical validity of those predictions, we need to investigate convergence. In our model fitting algorithm using MCMC, some signs of good mixing are:

- "Thick pen" behaviour in trace plots, indicating fast and efficient exploration of the target space
- Autocorrelation function which converges to 0 when increasing lag
- High effective sample size, a measure indicating how many independent samples our generated samples are "worth"

4.1 Implementation of MCMC trace diagnostics

To better understand convergence in our model fits, and allow users to do the same, it was necessary to add new functionality to the AddiVortes code. As mentioned before, Python was selected for further use and hence a new function allowing the user to view convergence diagnostics plots was added to the py-addivortes package [4].

The new `trace_diagnostics` function gives the user the effective sample size, as well as trace, histogram, and autocorrelation plots for the following 3 stats:

- The average number of centers per tessellation
- The average number of dimensions per tessellation
- Sigma, representing the noise in the system

The example below shows the outputs of this function relating to the average number of dimensions:

Output

```
Effective sample size (Average Number of Dimensions): 1839.7
Press [Enter] to see trace for Average Number of Dimensions:
Press [Enter] to see histogram for Average Number of Dimensions:
Press [Enter] to see autocorrelation for Average Number of Dimensions:
```

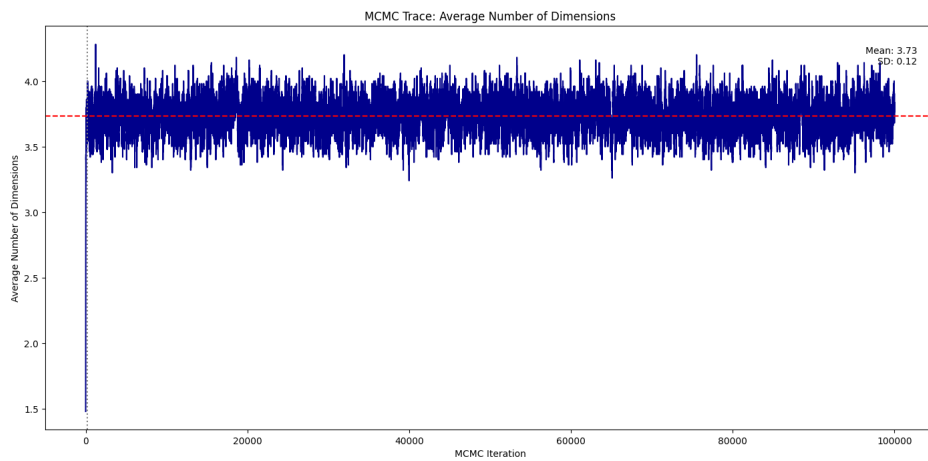


Figure 1: Trace plot for average number of dimensions

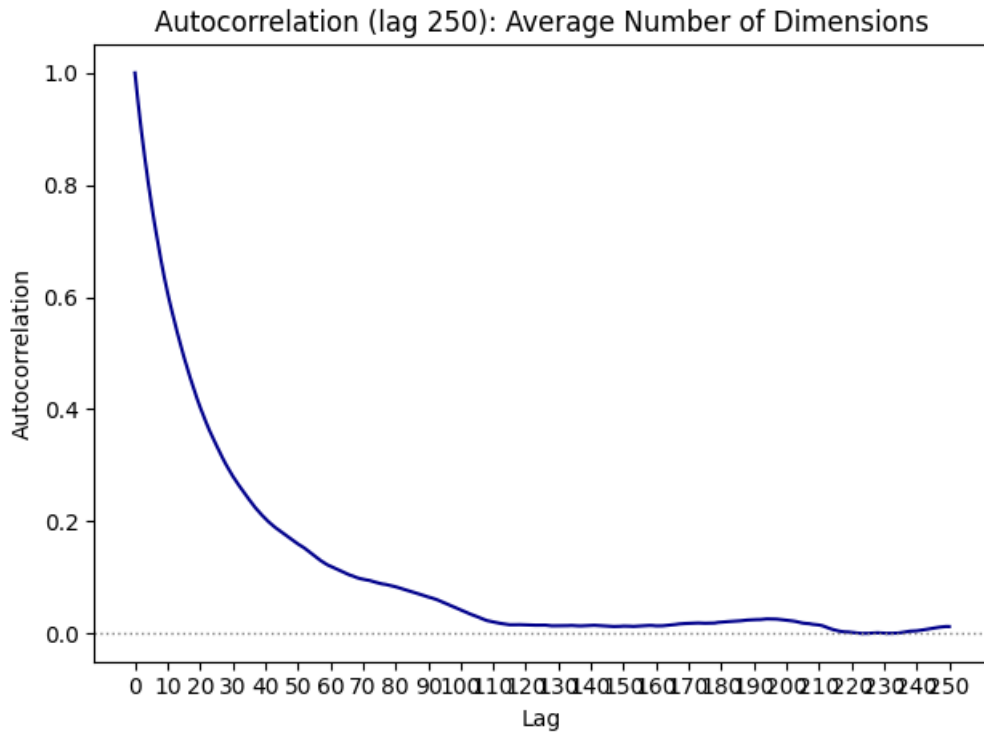


Figure 2: Autocorrelation plot for average number of dimensions

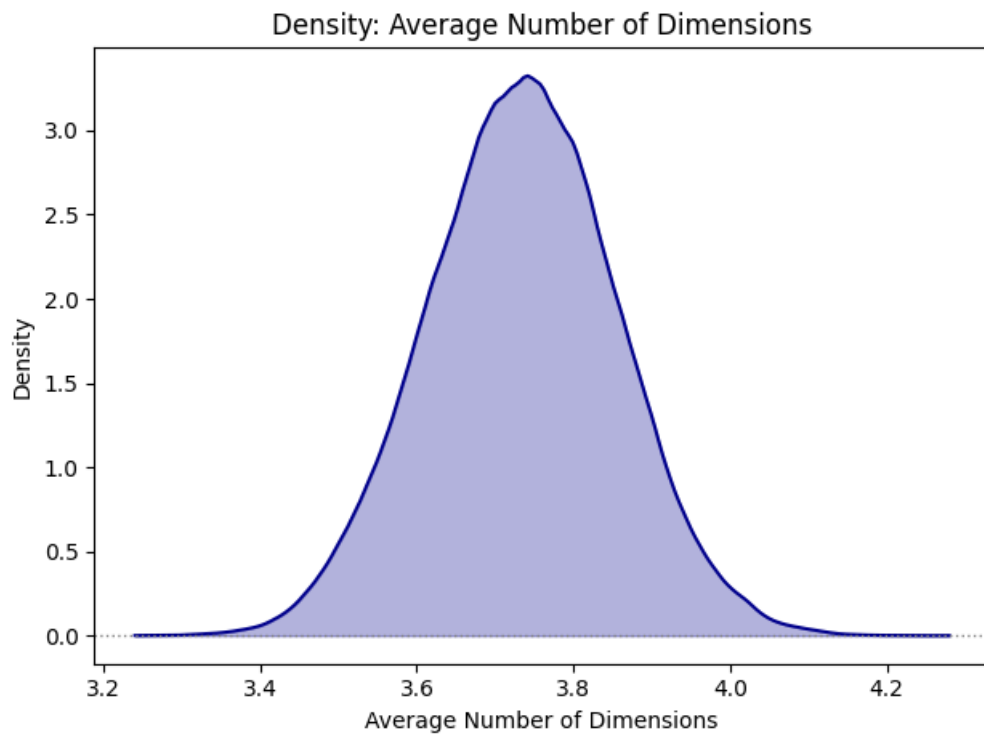


Figure 3: Histogram for average number of dimensions

5 Earthquakes dataset

In order to carry out further testing, we would need a well-produced large-scale dataset with a large number of covariates and observations. We were looking for something that used a variety of metrics, and had variables on varying scales. For that reason we decided to use the earthquakes dataset for further testing. The relevant files for this chapter can be found at `benchmarks\datasets\earthquakes` [2]

5.1 Data Source

The raw data was obtained from the U.S. Geological Survey’s Earthquake Hazards Program, via the public ANSS Comprehensive Earthquake Catalog (ComCat) query API [5].

The pull used for this article consists of the most recent 20,000 earthquake records in the catalog which occurred in 2025 (magnitude ≥ 2.5), which corresponds to events occurring between 5 May 2025 and 31 December 2025. The USGS API caps any single query at 20,000 returned rows; a full multi-decade pull requires paginating over the `offset` parameter or splitting the request into shorter date ranges, but the analysis in this article is restricted to this single 20,000-row pull as a representative working sample.

The raw file, `raw_data.csv` [2], contained 20,000 rows and 19 columns: `time`, `latitude`, `longitude`, `depth`, `mag`, `magType`, `nst`, `gap`, `dmin`, `rms`, `net`, `type`, `horizontalError`, `depthError`, `magError`, `magNst`, `status`, `locationSource`, and `magSource`.

5.2 Data Cleaning

The raw data required six cleaning steps: (i) restricting to genuine earthquake events, since the catalog also contains a small number of mining explosions and other non-tectonic events; (ii) removing columns that were pure metadata, near-duplicates of another retained column, or deliberately excluded from the benchmark covariate set; (iii) removing the small proportion of rows with residual missing values in the quality-control columns; (iv) subsetting to reviewed body-wave magnitude events, i.e. `magType == "mb"` and `status == "reviewed"`; and (v) dropping the now-constant `magType`, `status`, and `net` columns from the retained covariate set. The full cleaning pipeline, implemented in R, can be found at `clean_earthquakes.R` [2]. Applying this pipeline reduced the raw extract from 20,000 rows and 19 columns to a final cleaned dataset of 11,697 rows and 11 columns, with zero remaining missing values. Table 4 summarises each stage.

Stage	Rows	Columns
Raw extract	20,000	19
After filtering to <code>type == "earthquake"</code>	19,841	19
After dropping redundant columns	19,841	14
After dropping incomplete rows	18,732	14
After subsetting to <code>magType == "mb"</code> and <code>status == "reviewed"</code>	11,697	14
After dropping constant <code>magType</code> , <code>status</code> , and <code>net</code> columns	11,697	11

Table 4: Row and column counts at each stage of the cleaning pipeline.

5.3 Covariate Descriptions

Table 5 describes each of the 11 covariates retained in the final cleaned dataset, along with its type and the range (for numeric variables) or levels (for categorical variables) observed in the data.

Covariate	Type	Description	Range / Levels
latitude	Numeric	Epicenter latitude in decimal degrees (WGS84). Natural great-circle covariate.	-65.6° to 87.1°
longitude	Numeric	Epicenter longitude in decimal degrees (WGS84). Natural great-circle covariate.	-180.0° to 180.0°
depth	Numeric	Depth of the hypocenter below the surface, in kilometers. Small negative values reflect very shallow events located slightly above the reference datum.	-3.4 to 669.6 km
mag	Numeric	Earthquake magnitude, on the body-wave scale used for the retained events. Natural target variable.	2.5 to 8.8
nst	Numeric	Number of seismic stations used to determine the event's <i>location</i> .	3 to 566
dmin	Numeric	Horizontal distance, in degrees, from the epicenter to the nearest reporting station.	0 to 55.9
rms	Numeric	Root-mean-square of the travel-time residuals, in seconds; a location goodness-of-fit measure.	0 to 2.0 s
horizontalError	Numeric	95% confidence radius on the epicenter location, in kilometers.	0 to 40.8 km
depthError	Numeric	95% confidence uncertainty on the depth estimate, in kilometers.	0 to 42.9 km
magError	Numeric	Standard error on the magnitude estimate, in magnitude units.	0 to 1.3
magNst	Numeric	Number of seismic stations used to compute the <i>magnitude</i> (distinct from <i>nst</i>).	0 to $1,027$

Table 5: Covariates in the cleaned earthquake dataset.

For the use of this dataset in testing, we used `mag`, the magnitude of the earthquake, as the response variable.

6 Investigative testing using HPC resources

For the sake of further testing, we made use of the Hamilton HPC service of Durham University. The code used to generate the following test results were written with the following properties in mind:

- Parallelism: With each compute node on Hamilton having 128 CPU cores, making effective use of parallel processing was crucial in speeding up tests
- Pausable: For long processes which we would expect to take over a day to run, we designed test code such that a log of which point in the code execution was reached, so that if a process

was interrupted, for example by a runtime error or system downtime, then the tests could be resumed from the same place.

The files referenced in this chapter can be found at `AddiVortes\testing` [2]

6.1 Testing sensitivity of key metrics to parameter changes

We wanted to better understand the impact of model parameter settings on two key metrics: model fit time, and out-of-sample RMSE. The important thing to try and observe here is whether there was a trade-off between these 2 values, and if so, at which point do the improvements in out-of-sample RMSE start to diminish?

We first designed code to generate a factorial grid of possible model parameters. This simply gave parameters equally spaced values over their search range, being able to specify how many values were needed in each parameter range. This was our first, naive approach to the problem of hyperparameter optimisation, with the result being an equally spaced grid over parameter space. Whilst this doesn't consider how metrics may be affected by different parameters in different ways, it would give us a clearer idea of how changing certain parameters may affect metrics more than others.

We considered the following parameters, which could be changed in the Python implementation: `m`, `nu`, `q`, `omega`, `lambda`, `iter`, `burnin`. The code used to generate the equally spaced factorial grid can be found at `grid.py` and the output can be found at `grid.csv` [2].

We then fit an AddiVortes model using each row of parameter settings from the grid, and recorded

m	nu	q	omega	lambda	iter	burnin
20	1	0.6	1	1	500	0.01
20	1	0.6	1	1	500	0.5
20	1	0.6	1	1	100000	0.01
20	1	0.6	1	1	100000	0.5
20	1	0.6	1	50	500	0.01
20	1	0.6	1	50	500	0.5
20	1	0.6	1	50	100000	0.01
20	1	0.6	1	50	100000	0.5
20	1	0.6	7	1	500	0.01
20	1	0.6	7	1	500	0.5

Table 6: The first 10 rows of the factorial grid with 2 options for each parameter

fit time, prediction time, and in-sample and out-of-sample RMSEs. The code used to run these tests is available at `grid_tests.py` and the results are available at `grid-testlog.csv`

TODO analysis of the partial results obtained.

6.2 Parameter screening

In order to determine which parameters have a non-linear effect on the metrics, these being the parameters whose effect on the metrics would have to be investigated more closely, we carried out screening. The method used is built upon the elementary effects (EE) method for screening and uses a threshold value to separate the inputs with linear and nonlinear effects [6]. The code used to implement this method can be found at `AddiVortes\screening`.

Here, One-at-a-time (OAT) moves are performed along each parameter axis from a set of starting points. From each point, we perform OAT moves and compute elementary effects for every input factor. In the following output graphs, μ^* is a measure of the average size of the effect generated by moves in a particular parameter, and σ is a measure of the spread of the sizes of the effects on

that parameter. If the value of σ falls under some value σ_0 then we determine it to have a linear or negligible effect, meaning the parameter's effect is sufficiently consistent across the input space to be considered approximately linear. Otherwise, we determine it to have a non-linear or interaction effect, meaning the parameter's effect changes substantially across the input space.

Because of the computational intensity of the screening algorithm used, mainly due to the large number of model fits required over varying areas of the input space, we used the Boston dataset, which is significantly smaller in size than the Earthquakes dataset we have been using for our other analysis. We carried out the screening on the model parameters: $m, \nu, q, \omega, \lambda$, to determine if their effects were linear in the following 4 metrics, treated as the output of the algorithm in each test: in-sample RMSE, out-of-sample RMSE, fit time, and predict time. The parameters were varied across the ranges given in Table 1. The results, showing the measures μ^* and σ are below. Points marked in blue has values of σ lower than the threshold σ_0 and were hence identified as linear/negligible, and points marked in red were marked as non-linear/interaction.

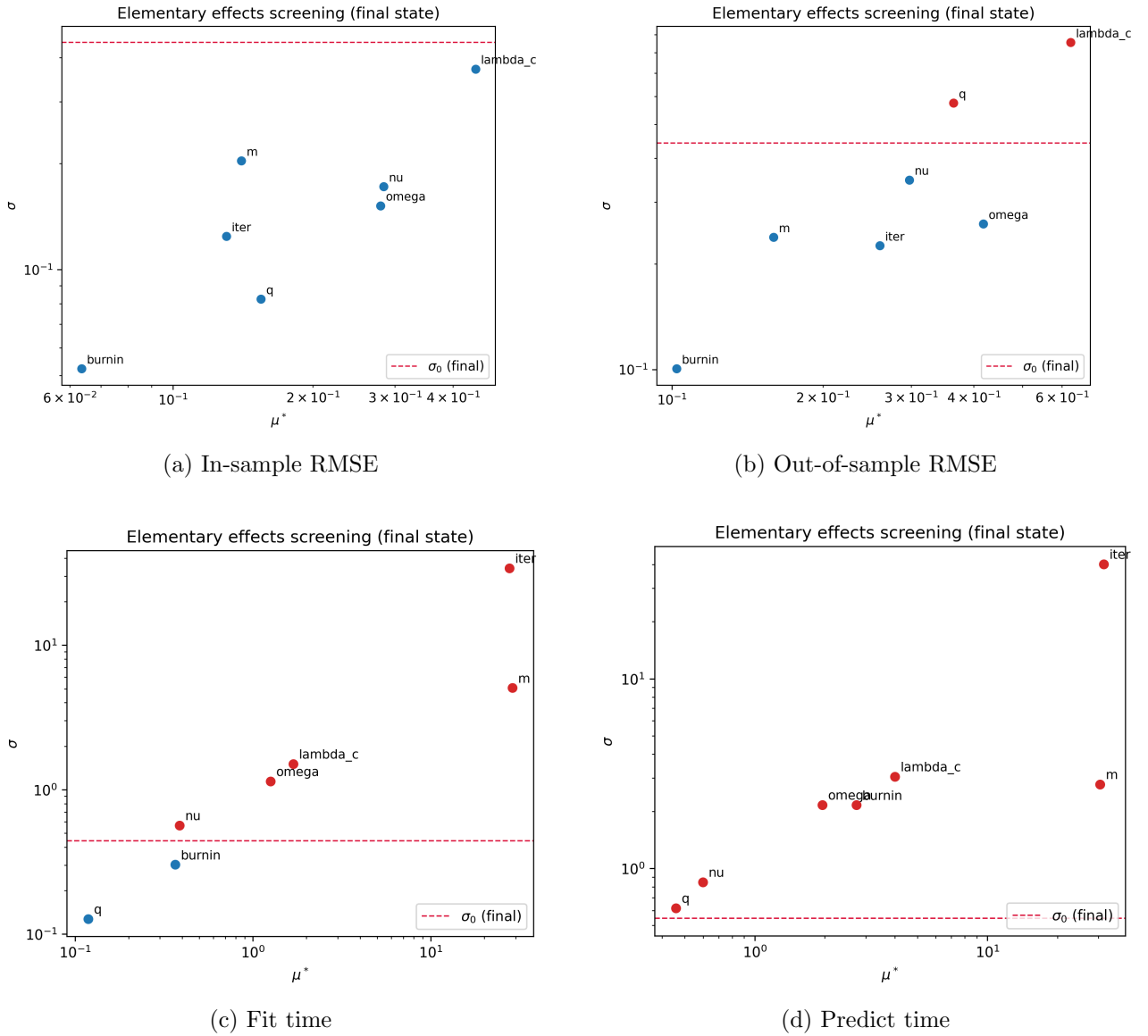


Figure 4: Screening results of model parameters on key metrics

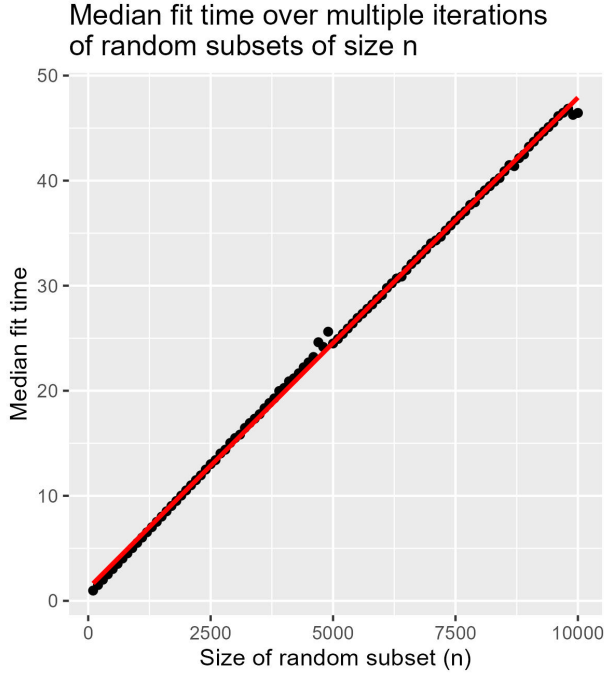
We will investigate the local effects of changing these parameters later on, and we can use the results of this screening to shape further investigation, and to determine if we can generalise results we obtain later.

6.3 Testing sensitivity of key metrics to change in dataset size

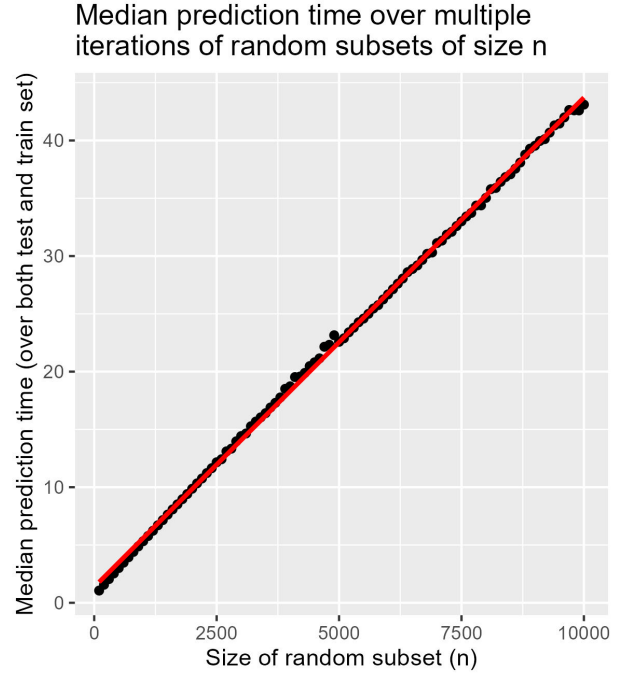
We wanted to explore how our key metrics, namely, in-sample, and out-of-sample RMSE, as well as our model fit and prediction times grow as the number of observations in the dataset (n) grows.

I wrote code to generate random subsets of the Earthquake dataset of sizes $n = 100, 200, \dots, 10000$. In order to obtain a broader picture of the trends as we vary n , each size, e.g. $n = 100$ was generated independently k times, taking the maximum possible value of k such that $nk \leq 100000$, to bound the time needed to generate and work on subsets of any given size, and make each batch size a suitable unit process for multiprocessing. This was implemented to minimise the random noise in the metrics as we take varying sub-populations of the full data. This leads to us taking $k = \lfloor \frac{100000}{n} \rfloor$. The code used to generate the subsets, as well as the subsets themselves for reproducibility, are available at `AddiVortes\effects\n` [2]

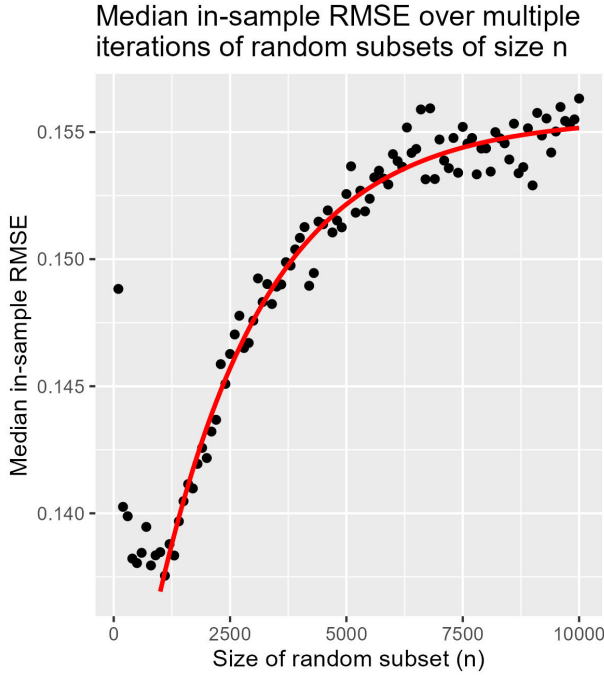
Once the subsets were generated, an AddiVortes fit was carried out on each subset, and the median statistics over each batch size were calculated. The results of which can be found at `AddiVortes\effects\n\addivortes_results.csv`. Graphs of the result summaries are below:



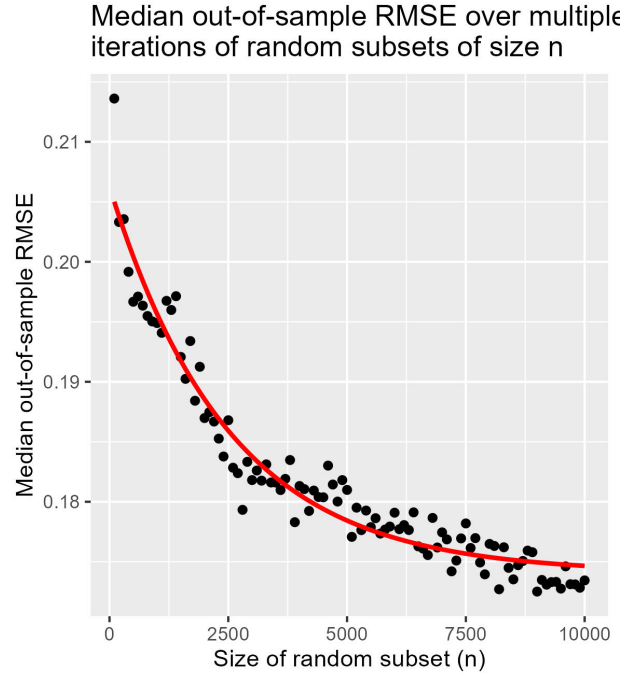
(a) Median fit time



(b) Median prediction time



(c) In-sample RMSE



(d) Out-of-sample RMSE

Figure 5: Performance metrics as a function of subset size n .

The red fit lines on the fit and prediction time graphs are linear, with small intercept coefficients (1.188 and 1.367 respectively for this data), which represent the overheads in the fit and prediction algorithms. Since the intercept term is small, when we double our number of observations n , we expect our prediction and fit time to also roughly double. These linear models fit the data well with adjusted R^2 values of 0.9992 and 0.9993 respectively.

The red fit lines for the in-sample and out-of-sample RMSEs instead use an asymptotic model of the following form:

$$y = a + b \exp(-nc) \quad (1)$$

This represents our theoretical understanding of RMSEs, with in-sample RMSE rising as we increase n , and the model has to be more flexible to fit a wider set of values, eventually flattening out, and out-of-sample RMSE falling as the wider set of data allows for a more flexible trained model which allows for greater predictive power on unseen data.

As visible in figure 4, in-sample RMSE starts off very high before taking a dramatic drop and then increasing steadily along towards the asymptote. This can be explained since for low values of n we would expect the random subsets to not be a representative sample of a much larger, and complex dataset such as the Earthquakes dataset. Hence, for our fit here we have ignored values of $n < 1000$ to allow for a better fit with this model. Both of these models were fit using non-linear least squares to estimate the optimal values of the parameters. This yielded adjusted R^2 values of 0.9742 and 0.9450 for in-sample and out-of-sample respectively.

In this model, if we double n , we expect the value of $y - a$ to increase by a factor depending on n .

$$y_0 = a + b \exp(-nc), y_1 = a + b \exp(-2nc) \quad (2)$$

$$\frac{y_1 - a}{y_0 - a} = \frac{b \exp(-2nc)}{b \exp(-nc)} = \exp(-nc) \propto \exp(-n) \quad (3)$$

Hence when we double our number of observations from a starting value of n , we only expect our out-of-sample RMSE to decrease by a factor proportional to $\exp(-n)$.

This clearly demonstrates that there will be some threshold of diminishing returns where increasing n will have a large effect on the fit and prediction time of the model but will have a very minimal effect on the out-of-sample RMSE, and hence at a certain value of n we want to stop increasing it, as we deem to have already gotten low enough out-of-sample RMSE.

References

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